

MOHAMMAD HADI NAJAFI

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Specialist in Embedded Systems and Product Development with over 7 years of proven track record in engineering firmware and hardware, and a history of leading cross-functional teams to deliver industrial-grade products. Award-winning engineer with international recognition in robotics, combining deep expertise in **PCB design**, **Industrial protocols**, and **Agile-Scrum** methodologies to drive technical innovation and product excellence.

EXPERIENCE

• Embedded Systems Engineer | [Naya, Netherlands](#) | September 2024 – Present

Developed a secure and reliable bootloader for the flagship product (Create), while deploying and integrating several new features into the firmware for the entire product line (Create, Touch, Track, and Tune).

SKILLS: Zephyr RTOS | C/C++ | BLE | Python | MCUBoot | STM32 | NRF52 | ARM Cortex-M4 | CMake | Github | Docker

• Product Innovation Manager | [Zarrin Samane Shargh Co., Mashhad.](#) | May 2022 – Dec 2023

Directed the end-to-end design and delivery of five new industrial products while managing a team of four engineers and earning recognition as the company's highest-rated unit manager.

Projects: DLMS Gateway | LTE Industrial Modem | Mid-Voltage Fault Detector | Device Tracker | Conference Audio System

SKILLS: C/C++ | Agile-Scrum | Embedded Linux | PCB design | APQP | Github | STM32 | IOT | GPS tracking | Cosem/DLMS | LTE | DNP3 | MODBUS | FOTA | Makefile | Altium | Digital Audio | Buildroot | Mediatek SOC | HTTP | UBoot

• Research and Development Engineer | [Zarrin Samane Shargh Co., Mashhad.](#) | April 2021 – May 2022

Refined and implemented cutting-edge embedded technologies into the product line, which led to the introduction of innovative projects and successfully resolved several commercialization challenges.

Projects: LTE Router | Outdoor Wireless Access Point | GPS Tracker

SKILLS: C/C++ | Embedded Linux | PCB design | Openwrt | IOT | LTE | GPS Tracking | FOTA | Altium | HTTP | Buildroot | Mediatek SOC | UBoot

• Embedded Systems Specialist | [Freelance](#) | March 2019 – April 2021

Designed and Delivered integrated hardware and firmware solutions for diverse clients, specializing in end-to-end product design.

Projects: AVL Tracking System | Multimedia Elevator Panel | NFC Access Control System | Visitor Audio Guide System | IoT for advertisement monitors

SKILLS: C/C++ | PCB design | C# | Python | IOT | GPS Tracking | STM32 | ESP32 | Altium | HTTP | Bare-metal Dev | Digital Audio | NFC | FreeRTOS

- **Software Developer** | [Kimia Control Co., Isfahan](#) | September 2018 – July 2019 (contract)

Co-developed an automated data-logging system to monitor precise quality parameters across all production stations, enabling real-time quality control and data-driven process optimization.

Projects: Smart Factory

SKILLS: LabVIEW | SQL | OPC | RS232

HONORS & AWARDS

- **3rd place in the RoboCup IranOpen competition** | [Tehran](#) | 2017

(@Work League) Optimized odometry precision and real-time computer vision performance.

- **Medallion for excellence in the 44th WorldSkills competition** | [Abu Dhabi](#) | 2017

(Mobile Robotics) Engineered high-precision navigation, Image processing and decision-making algorithms.

- **Bronze Medal in the 16th national skills competition** | [Tehran](#) | 2016

Enhanced robot kinematic precision and engineered a bespoke robotic arm tailored for project's requirements.

- **2nd place in the SharifCup robotics competition** | [Sharif University of Technology](#) | 2015

(Advanced path finder league) Implemented color detection and adaptive algorithm selection to optimize performance.

- **First place in 15th NadCup national competitions** | [Sharif University of Technology](#) | 2014

(Magnetic path finder robots league) Reduced the size of the magnet coils and increased their accuracy.

- **First place in 14th NadCup national competitions** | [Sharif University of Technology](#) | 2013

(Pathfinder robots league) Increased the number of sensors and reduced the robot's weight.

EDUCATION

- **B.Sc. in Electrical Engineering** | [Quchan University of Technology](#) | 2014 – 2020

Directed the University Drone & Robotics Research Group (2016 -2019).

Named "Best Researcher of the Year" (2015), recognized for outstanding technical contributions in robotics.

- **High School Diploma in Mathematics and Physics** | [Sampad \(NODET\)](#) | 2010 – 2014

(NODET): National Organization for Development of Exceptional Talents.